

Blood Pressure Monitoring System

1. Introduction

Over the past few days, I have been studying different approaches used for cuffless blood pressure monitoring. My main focus was to understand how existing systems work, the technologies involved, and the challenges that need to be solved while developing a wearable blood pressure monitoring device.

2. Study of Hilo Blood Pressure Monitoring System

Hilo is a wearable blood pressure monitoring system designed to estimate blood pressure continuously without using a traditional inflatable cuff.

Working Principle

The Hilo system primarily uses Photoplethysmography (PPG) signals obtained from the wrist. The sensor continuously measures blood volume changes in the arteries using light-based sensing.

The acquired PPG waveform is processed using Pulse Wave Analysis (PWA) and machine learning algorithms to estimate blood pressure.

Process Flow

Blood Flow Changes
↓
PPG Signal Acquisition
↓
Pulse Wave Analysis
↓
Machine Learning Model(calibration)
↓
Blood Pressure Estimation

Advantages

- Continuous monitoring
 - Comfortable wearable design
 - No inflatable cuff required
 - Suitable for long-term monitoring
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3. Electrocardiogram (ECG)

Principle

ECG measures the electrical activity generated by the heart during each cardiac cycle.

When the heart contracts and relaxes, electrical impulses are generated and travel throughout the body. These electrical signals can be detected at the skin surface.

ECG Waveform

The ECG waveform consists of:

- P Wave
- QRS Complex
- T Wave

The R-Peak present in the QRS complex is the most significant feature for timing analysis.

Importance in Blood Pressure Monitoring

ECG provides the exact moment when the heart initiates blood pumping. This information is used as the starting reference point for Pulse Transit Time calculations.

4. Photoplethysmography (PPG)

Principle

PPG is an optical sensing technique used to detect blood volume changes inside blood vessels.

The sensor consists of:

- LED
- Photodiode

The LED emits light into the skin.

A portion of the light is absorbed by blood while the remaining light is reflected back and detected by the photodiode.

As blood volume changes with each heartbeat, the amount of reflected light also changes, producing a PPG waveform.

Importance in Blood Pressure Monitoring

PPG detects the arrival of the pulse wave at the wrist and provides timing information required for blood pressure estimation.

5. Pulse Transit Time (PTT)

Definition

Pulse Transit Time is the time required for the pulse wave generated by the heart to travel from the heart to a peripheral location such as the wrist.

Calculation

$PTT = \text{Pulse Arrival Time} - \text{ECG R-Peak Time}$

Relationship with Blood Pressure

Higher Blood Pressure:

- Arteries become stiffer
- Pulse wave travels faster
- PTT decreases

Lower Blood Pressure:

- Arteries become more compliant
- Pulse wave travels slower
- PTT increases

Therefore, PTT has an inverse relationship with blood pressure.

Importance

PTT provides physiological information related to arterial stiffness and blood pressure and can be used as an additional feature for cuffless BP estimation.

6. Study of MAX86150 Sensor

Overview

The MAX86150 is an integrated biosensor designed for wearable health monitoring applications.

The sensor combines:

- ECG Acquisition
- PPG Acquisition
- Analog Front-End
- ADC Conversion

within a single device.

Key Features

- Simultaneous ECG and PPG acquisition
- Low power consumption
- Wearable device compatibility
- Compact form factor

Understanding

The MAX86150 is not a direct blood pressure sensor.

It acquires ECG and PPG physiological signals, which are later processed through algorithms to estimate blood pressure.

Challenges Identified

Motion Artifacts

- Movement affects ECG and PPG signals.
- Can reduce BP estimation accuracy.

Continuous ECG Acquisition

- Obtaining reliable ECG signals continuously from a wrist-worn device is challenging.

Calibration Dependency

- BP estimation accuracy depends on proper calibration.

Signal Quality

- Accurate BP estimation requires high-quality ECG and PPG signals.

7. Conclusion

The study focused on understanding the fundamental technologies involved in cuffless blood pressure monitoring. Existing systems such as Hilo mainly rely on PPG-based pulse wave analysis, while ECG, PPG, and Pulse Transit Time provide additional physiological information that can potentially improve blood pressure estimation. The MAX86150 sensor was studied as a key component for acquiring ECG and PPG signals required for advanced BP monitoring systems.